

TherMOS: A Toolkit for Low Resolution Thermal Modeling

DR. PASCAL SCHIRMER

Version History

#	Version	Date	Comment	Editor	Lead
1	v.1.0	15.05.2025	Creating initial version.	Pascal Schirmer	Pascal Schirmer
2	v.1.1	19.05.2025	Reviewed initial version.	Peter Schreivogel	Pascal Schirmer

Content

- 1) Short Introduction
- 2) Theoretical Background
- 3) Toolkit Overview
- 4) Exemplary Calculation
- 5) Advanced Functionalities
- 6) Outlook and Conclusion



A

SHORT INTRODUCTION

Aims and Goals

This MATLAB script/toolkit focuses on the reduction of high-order thermal models to more manageable lower-order models while preserving essential thermal dynamics. Thermal model order reduction is critical in simulations and real-time control applications where computational efficiency and accuracy are paramount. This process is typically necessary for systems with many thermal nodes, where full-scale modeling would be computationally prohibitive.

Dependencies

USED HARDWARE

- OS: WIN 10
- CPU: AMD 3700X
- RAM: 32 GB 3600 MHZ DDR4
- ROM: 1 TB SSD
- GPU: NVIDIA RTX 3700

SOFTWARE

The TherMOS Toolkit was implemented using Matlab R2023b and the following toolbox:

- distrib_computing_toolbox
- identification_toolbox
- image_toolbox
- matlab
- neural_network_toolbox
- optimization_toolbox
- statistics_toolbox

Limitations and Known Issues

- Numerical discrepancies for default implementation for calculating G_{th} and C_{th} matrices for 2D POD method
- Thermal structure function has convergence issues (should currently not be used)
- Deep learning does not support sequence-to-sequence learning



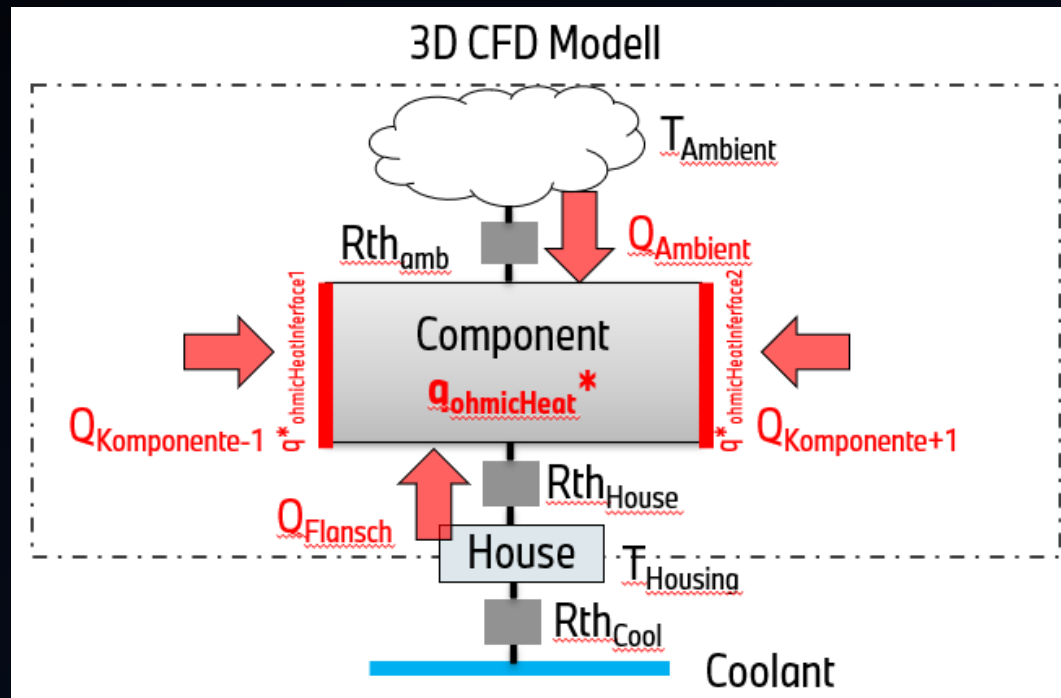
B

THEORETICAL BACKGROUND



Physical Model

Literature:	-
Images:	-
Software:	-
Example:	-
Content:	Fundamentals



Description

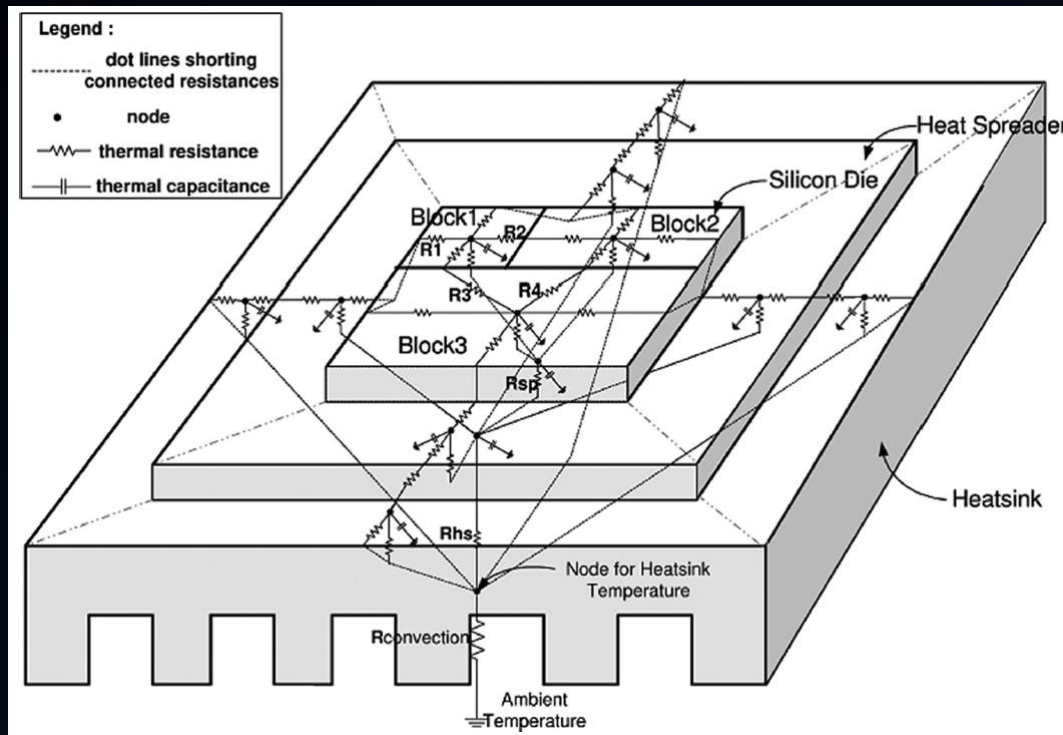
The generalized architecture that should be modelled using the ROMs is displayed below. It aims to represent a generalized component in a power electronic system that is dissipating losses, while having boundary conditions on system level. The following aspects are considered:

- 1) Internal heat generation ($q_{ohmicHeat}^*$)
- 2) External heating due to nearby sources ($q_{ohmicHeatInt}^*$)
- 3) Heat flow to the ambient (T_{amb})
- 4) Heat flow to the coolant ($T_{Coolant}$) over the housing ($T_{housing}$)

Therefore, the full order model consists next, to the geometry, of two independent reference temperatures and losses.

Reduced Order Model

Literature:	[5,6]
Images:	-
Software:	-
Example:	-
Content:	Fundamentals



Theory

The aim of a Reduced Order Model (ROM) is to reduce a complex geometrical physical model with high computational load using mathematical techniques. For example the temperature dependencies could then be modeled according to Fourier theory as a spatial temporal variable with infinite number of modes [2]:

$$T(x, y, t) = \sum_{i=1}^{\infty} \theta_i(t) \varphi_i(x, y)$$

where $T(x, y, t)$ is the spatial temporal temperature constructed by an orthonormal set of spatial vectors $\varphi(x, y)$ and corresponding temporal modes $\theta(t)$. The model coefficients are optimized to fulfil a minimum criteria, as:

$$\min_{\varphi_i(x, y)} \|T(x, y, t) - \hat{T}(x, y, t)\|$$



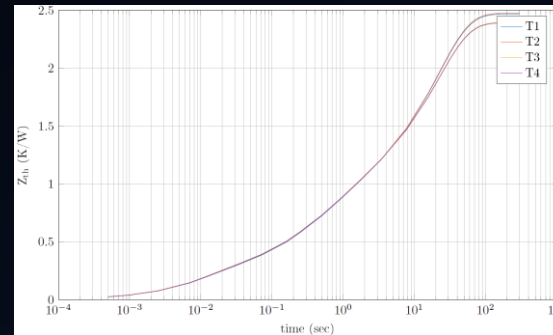
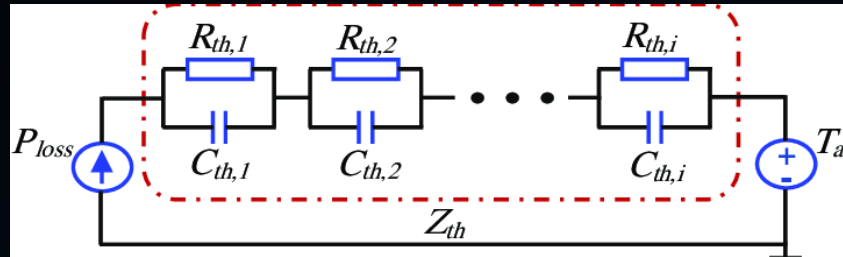
B1

1D REDUCED ORDER MODELS



Foster- and Cauer-models (RC-Models)

Literature:	[5,6]
Images:	-
Software:	-
Example:	-
Content:	Fundamentals



Example

Complex thermal resistance $Z_{th} \left(\frac{\text{K}}{\text{W}} \right)$ for four different components with different thermal coupling.

Theory

Foster and Cauer models (RC-Models) are one of the easiest form to fit reduced order thermal models based on step responses. The equation for fitting can be found below:

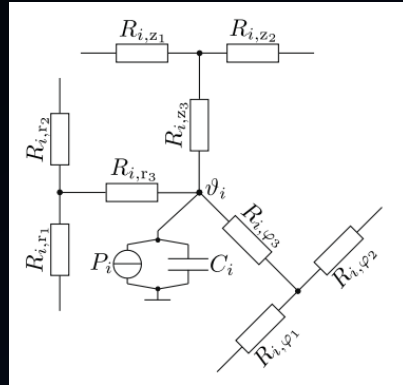
$$Z_{th}(t) = \sum_{i=1}^M R_{th}^i \left(1 - \exp \left(-\frac{t}{R_{th}^i C_{th}^i} \right) \right)$$

where Z_{th} is the fitted step-response for the M^{th} order model, and R_{th}^i / C_{th}^i are the i^{th} element of the thermal resistance and thermal capacitance respectively. The dynamic time discrete temperature $T_j[k]$ can then be written as:

$$T_j[k] = \sum_{i=1}^M \left(\frac{2\tau_i - \Delta t}{2\tau_i + \Delta t} T_i[k-1] + \frac{R_{th}^i \Delta t}{2\tau_i + \Delta t} (p_L[k] + p_L[k-1]) \right)$$

State-Space Models (SS-Models)

Literature:	[7]
Images:	-
Software:	-
Example:	-
Content:	Fundamentals



Example

Assuming that parameters are constant during the sampling interval state-space models can be discretized:

$$x[k+1] = \Phi x[k] + Hu[k] \text{ and } y[k] = Cx[k]$$

$$\Phi = e^{AT_s}$$

$$H = \int_{t_k}^{t_{k+1}} e^{A\tau_s} d\tau_s B$$

Theory

After carrying out all network simplifications by parallel and series connections of thermal resistors, each LPTN topology can be rewritten in the form of ordinary differential equations [3]:

$$C_i \frac{\partial \vartheta_i}{\partial t} = P_i(t) + \sum_{j=1}^n \frac{\vartheta_j - \vartheta_i}{R_{ij}^{th}} + \sum_{k=1}^m \frac{\vartheta_{ref,k} - \vartheta_i}{R_{ik}^{th}}$$

where R_{ij}^{th} are the internal nodes and R_{ik}^{th} are the nodes to the reference temperature. Rearranging the above equations leads to the following state-space model:

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t)$$

where x represents the nodal temperatures ϑ and u is the input vector consisting of power losses P and the reference temperatures ϑ_{ref} .

Parameter varying Models

Literature:	[5]
Images:	-
Software:	-
Example:	-
Content:	Fundamentals

Approximation

For a digital implementation on embedded hardware, the LPTN state-space model has to be discretized. Assuming that the input and scheduling variables are constant during a sampling interval:

$$\begin{aligned}x[k+1] &= \Phi(\zeta(k))x[k] + H(\zeta(k))u[k] \\ y[k] &= Cx[k]\end{aligned}$$

$$\Phi(\zeta(k)) = e^{A(\zeta(k))T_s}$$

$$H(\zeta(k)) = \int_{t_k}^{t_{k+1}} e^{A(\zeta(k))T_s} d\tau_s B(\zeta(k))$$

Since A, B are non-constant matrices they have to be updated, thus approximations like Euler are used:

$$\begin{aligned}\Phi(\zeta(k)) &\approx I + T_s A(\zeta(k)) \\ H(\zeta(k)) &\approx T_s B(\zeta(k))\end{aligned}$$

Theory

Assuming that not all parameters are constant, the above models can be extended by a so-called scheduling variable $\zeta(t)$, indicating that parameters are non-constant [5]:

$$C_i(\zeta(t)) \frac{\partial \vartheta_i}{\partial t} = P_i(t) + \sum_{j=1}^n \frac{\vartheta_j - \vartheta_i}{R_{ij}^{th}(\zeta(t))} + \sum_{k=1}^m \frac{\vartheta_{ref,k} - \vartheta_i}{R_{ik}^{th}(\zeta(t))}$$

Rearranging the above equations leads to the following state-space model:

$$\begin{aligned}\dot{x}(t) &= A(\zeta(t))x(t) + B(\zeta(t))u(t) \\ y(t) &= Cx(t)\end{aligned}$$

Feedback models

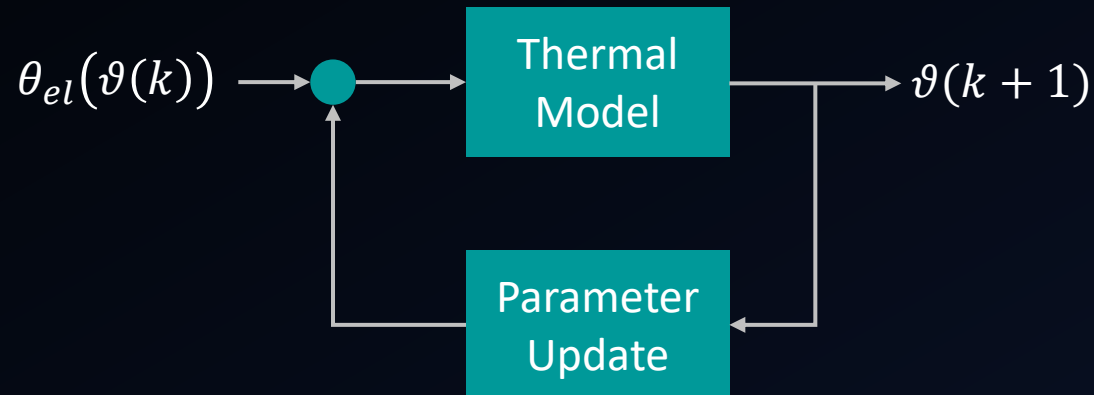
Literature:	-
Images:	-
Software:	-
Example:	-
Content:	Fundamentals

Theory

Often parameters are also sensitive to temperature, which leads to feedback models since the output influences the input of the model. For most cases it is sufficient to assume a linear relationship:

$$\theta_{el}(\vartheta) = [1 + \alpha(\vartheta - \vartheta_0)]\theta_{el}(\vartheta_0)$$

where $\theta_{el}(\vartheta)$ is a set of temperature dependent electrical parameters and ϑ_0 is the reference temperature. The input and output relationship can then be represented as indicated on the left. It must be noted that the feedback can lead to instable models.





B2

2D REDUCED ORDER MODELS



Proper Orthogonal Decomposition

Literature:	-
Images:	-
Software:	-
Example:	-
Content:	Fundamentals

- POD extracts dominant patterns (modes) from simulation data (snapshots).
- Modes capture most of the system's energy (variance).
- Automatic capturing of boundary conditions
- Capturing of physical properties in reduced model

$$T(x, y, t) = T_0(x, y) + \sum_{i=1}^K b_i(t) \cdot \varphi_i(x, y)$$

where T_0 is the time average of temperature (i.e., the mean vector of the observation matrix), $\varphi_i(x, y)$ is the i -th POD mode, and $b_i(t)$ is the temporal mode.

$$\sum_j^K c_{ij} \frac{db_j}{dt} + g_{ij} b_j = P_j(t)$$

where c_{ij} and g_{ij} are the elements of the thermal capacitance C_{th} and conductance G_{th} matrix, and $P_j(t)$ is the power loss source projected onto the POD space.

$$c_{ij} = \int_{\Omega} \rho c_p \varphi_i \varphi_j d\Omega$$
$$g_{ij} = \int_{\Omega} k \nabla \varphi_i \nabla \varphi_j d\Omega$$
$$P_j = \int_{\Omega} \varphi_j q_v(x, y, t) d\Omega$$



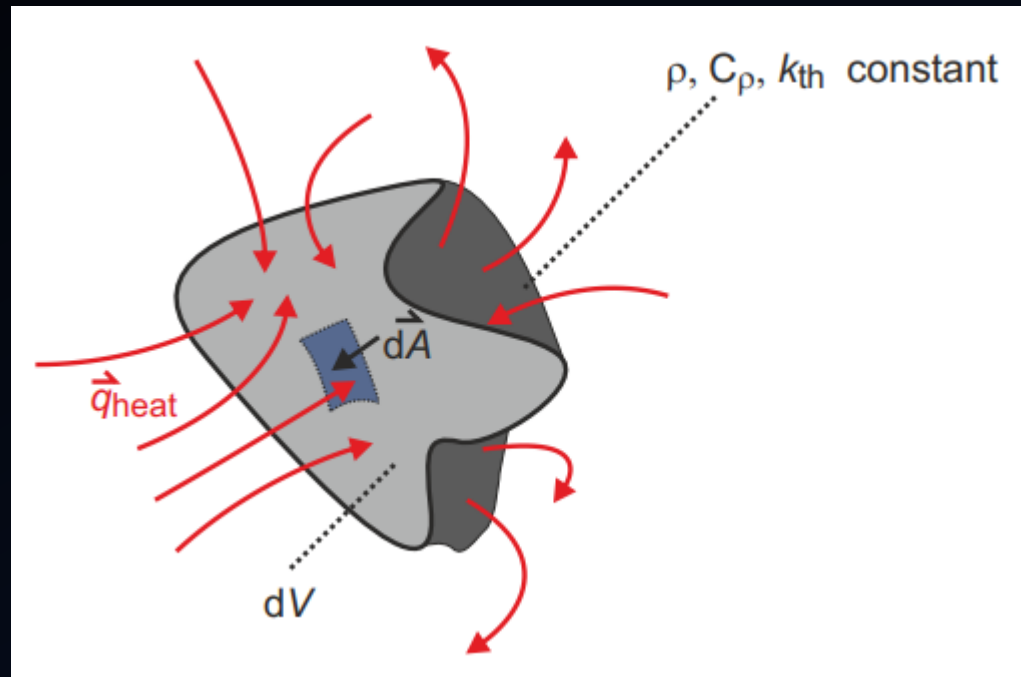
B3

PHYSICS INFORMED MODELS



Physically motivated background: General

Literature:	[8]
Images:	-
Software:	-
Example:	-
Content:	Fundamentals



Theory

To model heat transfer between bodies, the definition of the energy balance is crucial.

$$E_{heat} = T \cdot c_p \cdot \rho \cdot V$$

where E_{heat} is the energy, T is the temperature, c_p is the heat capacitance, ρ is the density, and V is the volume. The energy can only change due to a heat flux Q_{heat} or internal heat generation G_{heat} .

$$\frac{\partial E_{heat}}{\partial t} = Q_{heat} + G_{heat}$$

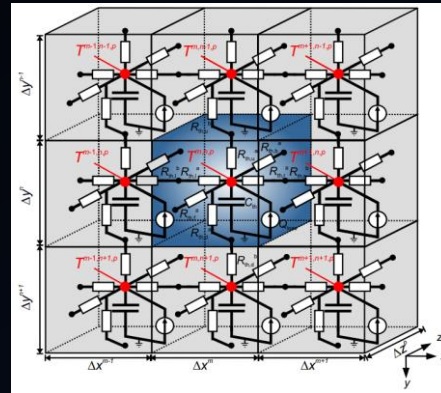
Applying Fourier's law the above equations can be rewritten into the typical heat equation.

$$\frac{\partial T}{\partial t} - \frac{k_{th}}{c_p \rho} \nabla^2 T = \frac{g_{heat}}{c_p \rho}$$

where k_{th} is the thermal conductivity and g_{heat} is the heat generation density.

Physically motivated background: Discretization

Literature:	[8]
Images:	-
Software:	-
Example:	-
Content:	Fundamentals



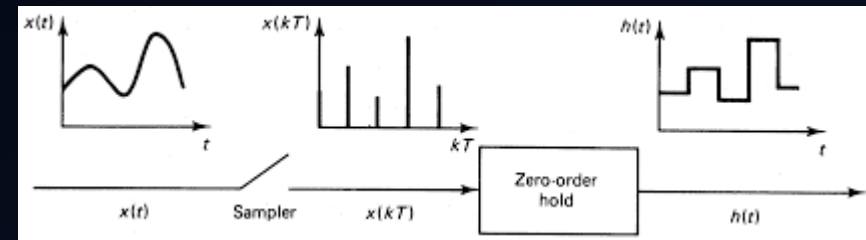
Spatial Discretization

Applying spatial discretization the heat equation can be rewritten as:

$$C_{th} \frac{dT}{dt} = \sum Q_{heat} + Q_{loss}$$

$$Q_{heat} = \frac{T_0 - T}{R_{th}}$$

where Q_{heat} is the heat flow and Q_{loss} are the internal losses.



Temporal Discretization

Using a thermal state-space model the results can be written as:

$$\frac{dT}{dt} = \mathbf{A} \cdot T + \mathbf{B} \cdot Q_{loss}$$

$$T(k+1) = \mathbf{A}^D \cdot T + \mathbf{B}^D \cdot Q_{loss}$$

For LTI systems and constant inputs $\mathbf{A}^D, \mathbf{B}^D$ can be written:

$$\mathbf{A}^D = \exp(\mathbf{A} \cdot T_s) \text{ and } \mathbf{B}^D = \int_0^{T_s} \mathbf{B} \cdot \exp(\mathbf{A} \cdot \tau) d\tau$$



C

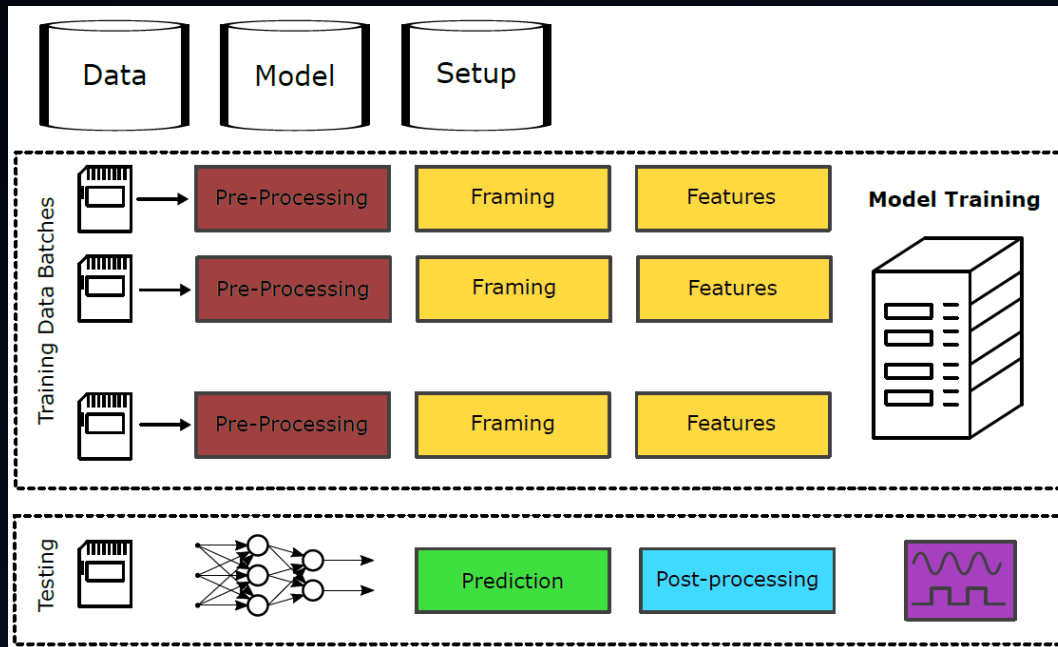
TOOLKIT OVERVIEW



Toolkit Overview

Literature:	-
Images:	-
Software:	TherMOS
Example:	-
Content:	Toolkit Description

OVERVIEW



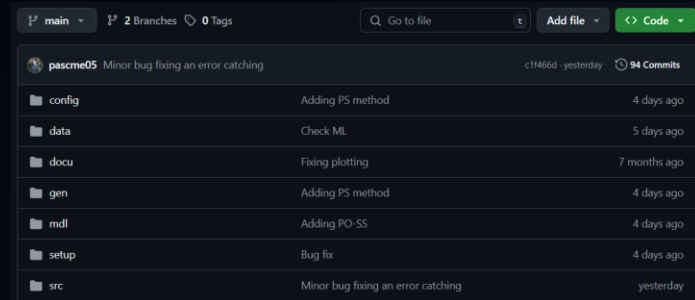
DESCRIPTION

- The toolkit implements model order reduction methods for 1D and 2D thermal problems
- The main contribution of the toolkit is the implementation of the proper orthogonal decomposition (POD) using a Galerkin and a state space approach
- 1D Foster networks and state space systems provide reference setups for simple approaches commonly used in industry
- Fundamentals of machine and deep learning are implemented as well

Installation and Usage

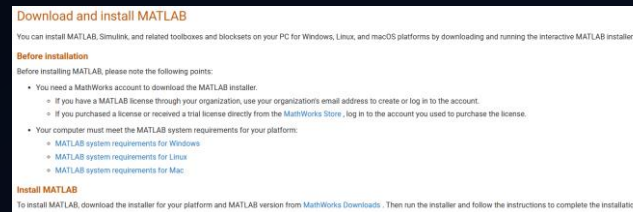
Literature:	-
Images:	-
Software:	TherMOS
Example:	-
Content:	Toolkit Description

STEP I: CHECK OUT GITHUB



Clone the latest version of the TherMOS Toolkit from Github:
<https://github.com/pascme05/TherMOS>

STEP II: INSTALL MATLAB



Install Matlab R2023b or higher to ensure the toolkit to run smoothly.

<https://de.mathworks.com/help/install/ug/install-products-with-internet-connection.html>

STEP III: RUN START.MAT

To execute a new simulation three things are needed. First, a valid configuration file. Second, a training and testing dataset. Third a start script as provided in start.mat used for defining the configuration and the datasets.

Project Structure

Literature:	-
Images:	-
Software:	TherMOS
Example:	-
Content:	Toolkit Description

TherMOS	Folder	Subfolder	Content
--	config		Contains the configuration files
--	data		Contains the datasets for training and testing
--	docu		Contains the documentation
--	gen		Contains code for generation of exemplary data
--	mdl		Contains trained models
--	results		Contains all results
--	setup		Contains preconfigured setup sheets
--	src		Contains source code implementation
	--	data	Contains functions for loading data
	--	dev	Contains functions for electrical calculation
	--	general	Contains general and helper functions
	--	model	Contains functions for reduced order models
	--	numeric	Contains functions for numerical integration
	--	plot	Contains functions for plotting

Setup File

Literature:	-
Images:	-
Software:	TherMOS
Example:	-
Content:	Toolkit Description

```
%% Simulation Setup
%%=====
% General
%%=====
setup.user = "Pascal Schirmer";
setup.name = "tutorial_SS";

%=====
% Experiment
%=====
% Configuration
%=====
setup.config = "tutorial_SS";

%-----
% Plotting and Saving
%-----
setup.plotOut = 1;
setup.plotMdl = 0;
setup.featRank = 0;
setup.save = 0;

%-----
% Training and Testing
%-----
setup.train = 1;
setup.test = 1;

%=====
% Data
%=====
% Format and Dimension
%-----
setup.format = 'xlsx';
setup.datDim = 1;
setup.datSep = 2;

%-----
% Training and Testing Selection
%-----
% File based
setup.trFile = ["data_Tutorial_1D"];
```

Description

- Experiment
 - Define training and testing
 - Define config file
- Data Files
 - Define data formats
 - Define inputs and outputs
- Models
 - Activate analytical 1D/2D models
 - Activate machine/deep learning

Literature:	-
Images:	-
Software:	TherMOS
Example:	-
Content:	Toolkit Description

Date	13.08.2024				
Author	Dr. Pascal A. Schirmer				
Version	v.0.0				
File	templateConfig				
#	Version	Date	Changes	Editor	Lead
1	v.0.0	13.08.2024	Initial implementation of the config template	Pascal Schirmer	Pascal Schirmer
2					
3					
4					
5					

Legend		
Color-Code	Meaning	Change
	Headings	No
	Information (can be changed for personal notes and markup)	Yes
	Input values	Yes
	Variable names read by toolkit	No

Name	Category	Description	Variable	Type	Value	Unit
Initialise Zero	gen	How the initial conditions of the output y are treated: 1) Unchanged 2) Set to zero 3) Remove reference value	init	integer+	2	-
Sampling Method	gen	Method for resampling time and spatial data: 1) Zero Order Hold (ZOH) 2) Linear Interpolated (LIN) 3) Spline Interpolation	samp	integer	1	-
Sampling Period	gen	Target sampling period for the internal data usage	Ts	float+	0.20	sec
Spatial Sampling X	gen	Spatial sampling in the first dimension x	dx	float+	0.01	m
Spatial Sampling Y	gen	Spatial sampling in the first dimension x	dy	float+	0.01	m
Pre-Padding	gen	Number of zero samples that are pre-padded to the input and output.	pad	integer+	0	-

LEGEND

- All changes to the templated are logged here
- New config files should be based on the default file

- Colour coding to indicate which cell should be changed and which not
- Colour code is valid throughout all worksheets

SETTINGS

- Four worksheets to adapt: experiment, data, models, and parameters
- Each worksheet as drop downs a limits for the values

Data Files

Literature:	-
Images:	-
Software:	TherMOS
Example:	-
Content:	Toolkit Description

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  
% Title: Thermal Model Order Reduction and Simulation (TherMOS) %  
% Topic: Power Electronics, Model Order Reduction %  
% File: data/readme %  
% Date: 18.08.2024 %  
% Author: Dr. Pascal A. Schirmer %  
% Version: V.0.1 %  
% Copyright: Pascal Schirmer %  
% Comments: %  
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  
  
TherMOS can take 1D or 2D/3D input data. For 1D either .xlsx or .mat files  
are available for 2D/3D only .mat files can be used. The following data  
structure must be followed:  
  
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  
%% Description 1D  
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  
%% Description 1D  
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  
The size of the data is Tx(M + 2N + 3) where T is the number of samples M  
is the number of input signals, N the number of output/reference signals.  
Three additional signals are needed for time, time per id, and the id  
label. The structure is as follows:  
  
1) data.time: Continuous time starting from 0 to (Ts*T-Ts)  
2) data.time_id: ID based time starting from 0 for each new ID  
3) data.id: ID label to separate several profiles within one file  
4) data.Fx: Input feature x  
5) data.Ox: Output temperature x  
6) data.Rx: Reference temperature x
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  
%% Description 2D  
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%  
The 2D/3D data file contains several variables, the variable name, meaning,  
size, and unit are listed below assuming a 2D problem with T temporal  
samples and M/N spatial samples:  
  
1) Cp: Spezific spatial heat capacitance of size MN (J/KgK)  
2) dx: Spatial resolution x direction (m)  
3) dy: Spatial resolution y direction (m)  
4) geo: Geometrical position in x,y,z of size MNx3 (m)  
5) k: Spatial thermal conductivity of size MN (W/mK)  
6) Lx: Length x direction (m)  
7) Ly: Length y direction (m)  
8) r: Spatial temperature reference of size TxMN (degC)  
9) rho: Spatial material density of size MN (kg/m3)  
10) t: Time vector of size Tx1 (sec)  
11) Ts: Sampling time (sec)  
12) X: Spatial input features of size TxMN (W)  
13) y: Spatial output features of size TxMN (degC)
```

README

- General overview about the dataset structure
- 1D data description
- 2D data description

1D MATLAB/EXCEL

- 1D timeseries can be defined in Excel or Matlab
- The format uses T samples, for M inputs and N outputs

2D MATLAB

- 2D spatial and temporal varying problems can be defined in Matlab only
- The format uses T samples and MN spatial values



D

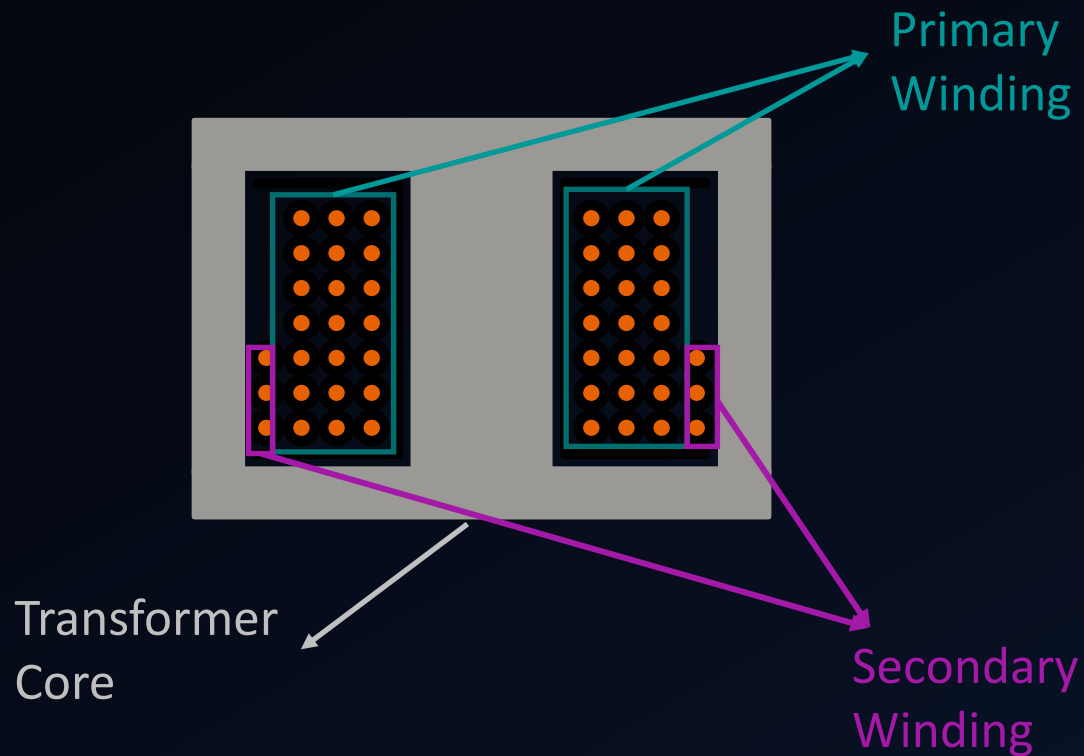
EXEMPLARY CALCULATIONS



Problem Description: 1D Dataset

Literature:	-
Images:	-
Software:	-
Example:	-
Content:	Dataset

PHYSICAL PROBLEM



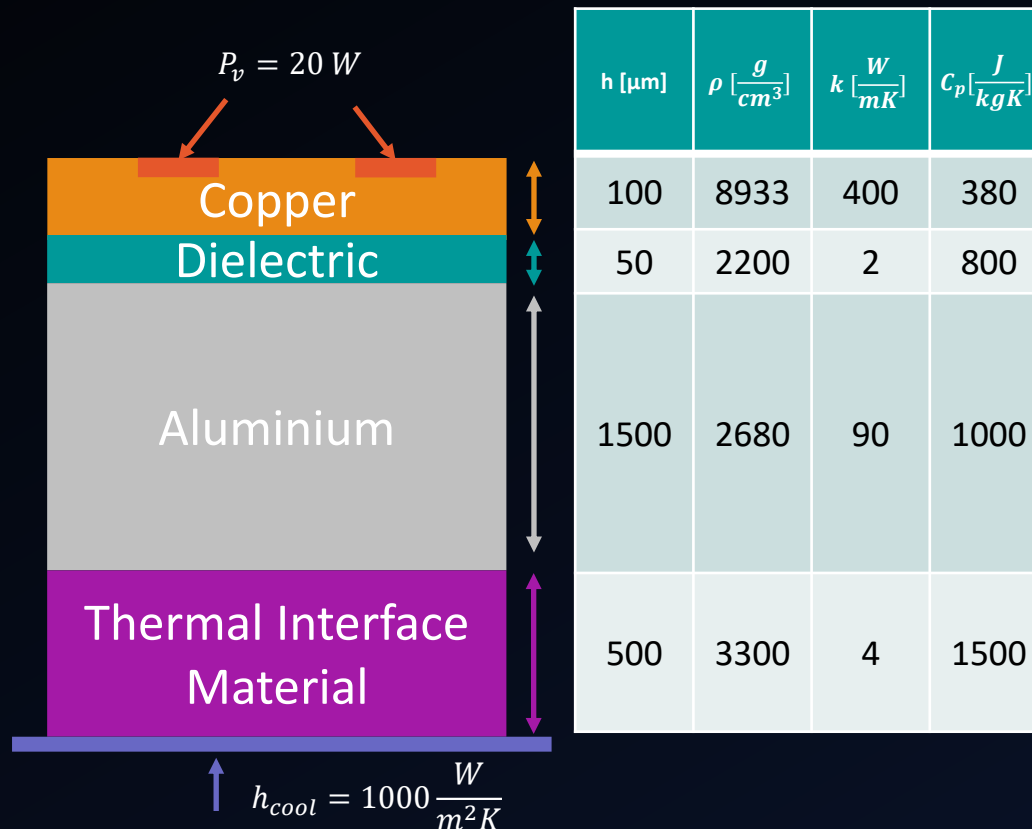
DATASET DESCRIPTION

- The data contains heat up curves for a transformer with E core
 - Input: Losses for primary, secondary, and core
 - Output: Temperatures for primary, secondary, and core
- The winding and the core are thermally coupled
- The amount of coupling depends on the operating point
- Three operating points are included (OP1-OP3)

Problem Description: 2D Dataset

Literature:	-
Images:	-
Software:	Matlab
Example:	journal3.m
Content:	Dataset

PHYSICAL PROBLEM



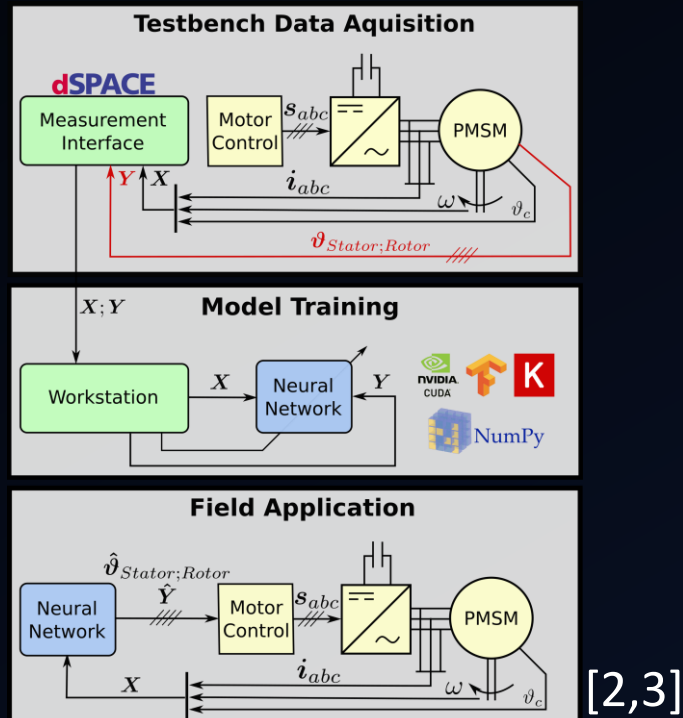
DATASET DESCRIPTION

- The dataset contains thermal data from an insulated metal substrate board (IMS) [1]
 - Losses from two power semiconductor switches
 - Convective boundary at the bottom
 - Adiabatic on all other edges
- The switches are coupled through the copper layer
- The dominant heat flux is in direction of the cooling

Problem Description: ML/DL Dataset

Literature:	-
Images:	-
Software:	-
Example:	-
Content:	Dataset

PHYSICAL PROBLEM



DATASET DESCRIPTION [2]

- The dataset contains ~180 hrs of measured testbench data for an electric machine (PSM)
 - Input: Currents, Voltages, torque, speed, etc.
 - Output: Stator and Rotor temperatures
- The dataset is cluster in IDs for splitting training, testing and validation
- Dataset made publicly available from Paderborn University: [4]

D1

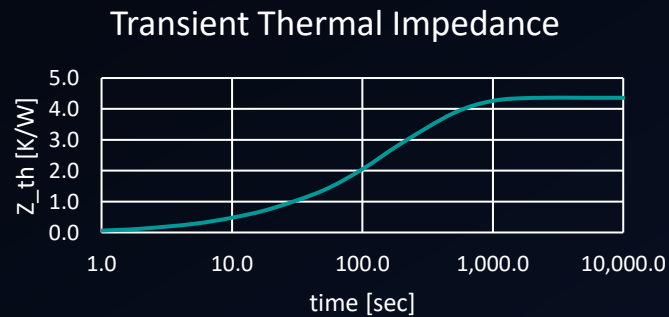
ANALYTICAL 1D MODEL ORDER REDUCTION

RC-Model: Training and Testing Results

Literature:	-
Images:	-
Software:	TherMOS
Example:	tutorial_RC.m
Content:	Tutorial

SETTINGS & PARAMETERS

- Fixed model order 3
- Foster structure
- Training OP1, Testing OP2

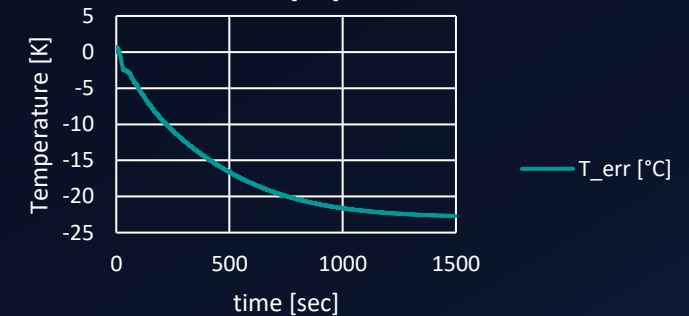
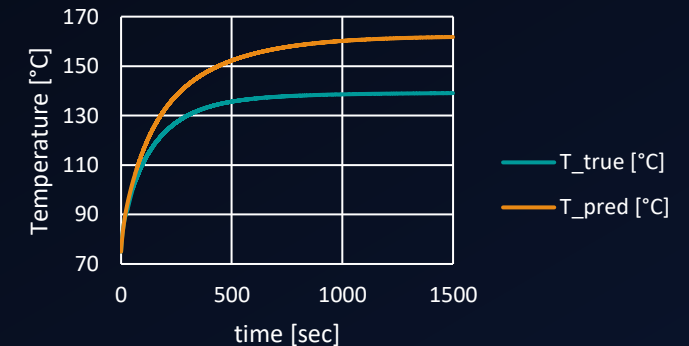


STEADY & TRANS. ERROR METRIC

	MAE	MSE	MAX
SS	18,55	364,5	22,70
TR	3,17	13,05	5,96
All	17,32	336,3	22,70

	Node-1	Node-2	Node-3
R_{th} [$\frac{K}{W}$]	1,60	2,40	0,36
C_{th} [$\frac{Ws}{K}$]	60,1	130,6	24,3

TESTING DATA PLOT



SS-Model: Training and Testing Results

Literature:	-
Images:	-
Software:	TherMOS
Example:	tutorial_SS.m
Content:	Tutorial

SETTINGS & PARAMETERS

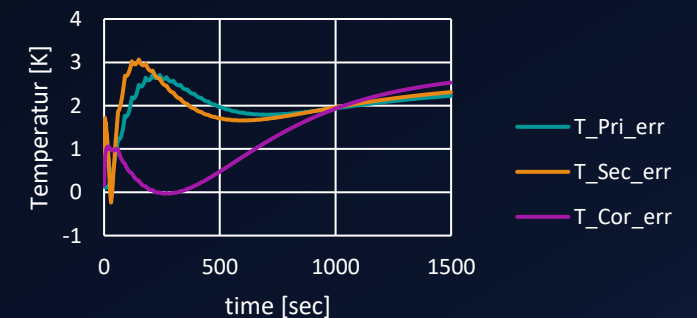
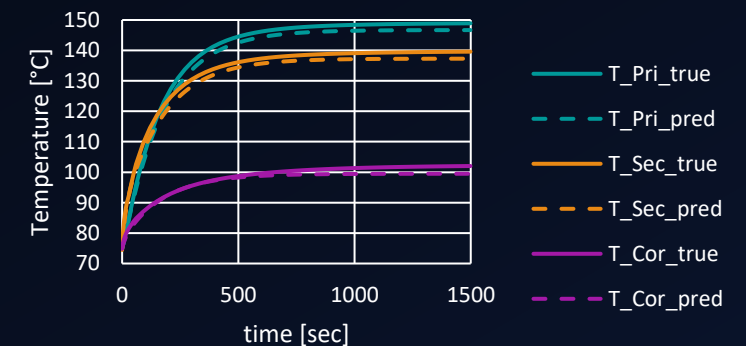
- Training: OP1/3, Testing: OP2
- Model Order: 2
- Inputs: Primary, secondary, and core losses
- Model form: Continuous
- Noise modeling: off
- Solver: linear

STEADY & TRANS. ERROR METRIC

SS	MAE	MSE	MAX
Pri	1,98	3,93	2,23
Sec	2,05	4,33	3,06
Core	1,35	2,62	2,53

TR	MAE	MSE	MAX
Pri	2,06	4,80	2,71
Sec	1,69	3,72	3,02
Core	0,71	0,58	1,06

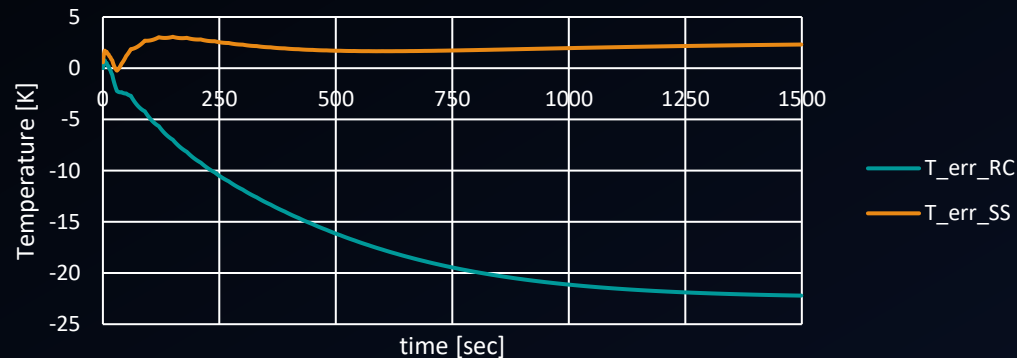
TESTING DATA PLOT



RC- and SS-Model Comparison

Literature:	-
Images:	-
Software:	TherMOS
Example:	tutorial_SS/RC.m
Content:	Tutorial

ERROR COMPARISON SECONDARY (OP2)



	MAE	MSE	MAX
SS	2,02	4,28	3,06
RC	17,32	336,3	22,70

DESCRIPTION & ANALYSIS

- State Space system outperform RC networks for MAE, MSE, and MAX error
- The losses in the transformer are coupled, thus primary, secondary and core losses mutually influence the temperature
- 1D RC networks (Foster) cannot predict the temperature accurately due to the coupling
- State Space models include the thermal coupling resistances and the power mapping inside their state space matrices



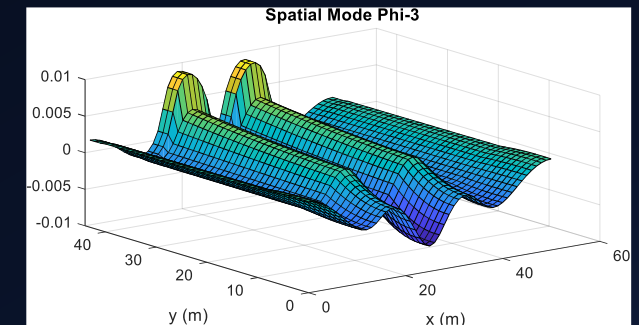
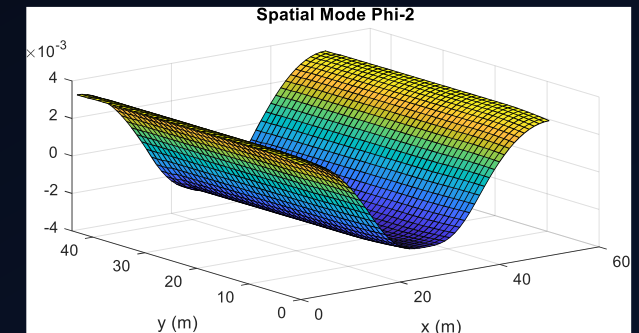
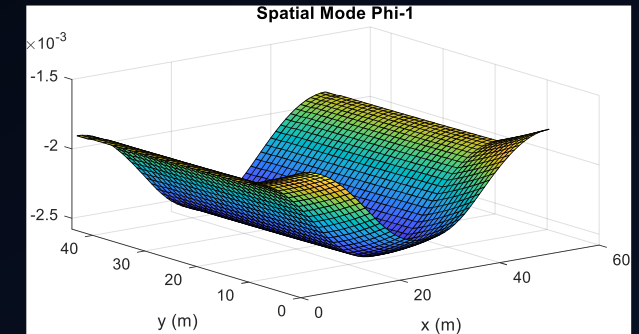
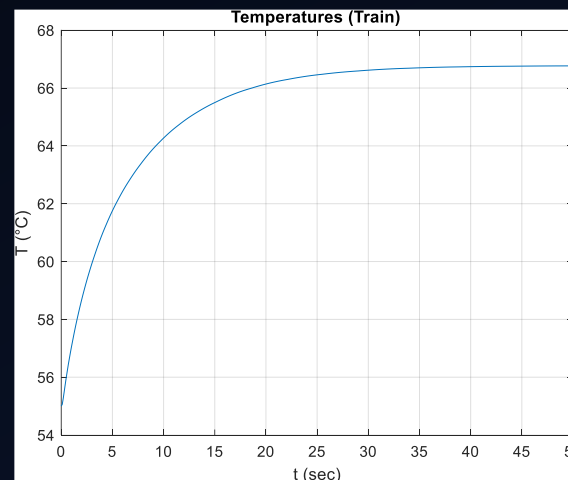
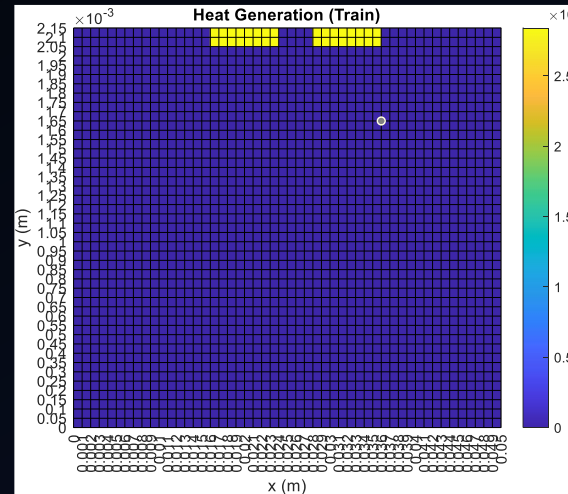
D2

ANALYTICAL 2D MODEL ORDER REDUCTION



POD-Model: Training Spatial Modes

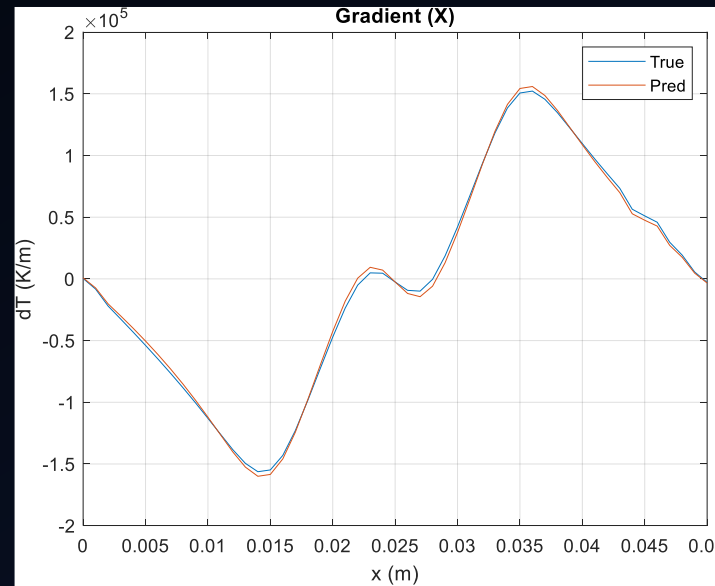
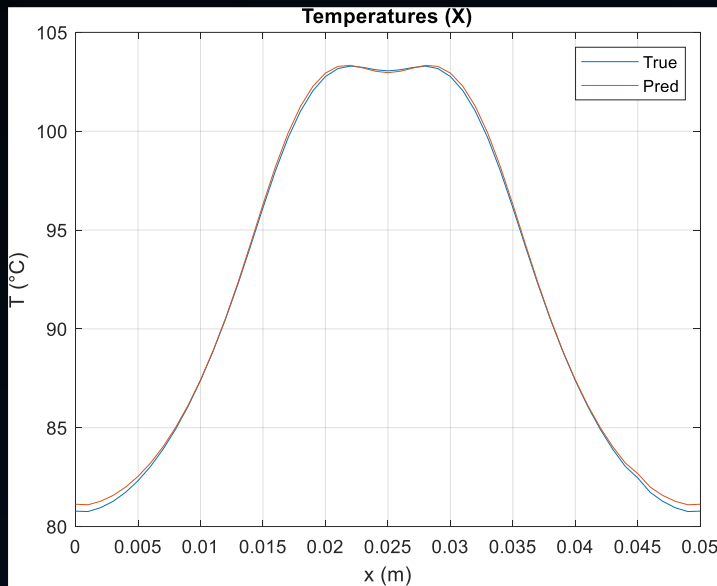
- Training based on one step response with 5 W losses ($2,84E8$ W/m³)
- Using fixed number of POD modes equal to three
- Using ode23s as stiff solver for Gth and Cth system
- Using Gaussian Legendre Integration and Jacobi matrix for mesh correction



Literature:	-
Images:	-
Software:	TherMOS
Example:	tutorial_PO.m
Content:	Tutorial

POD-Model: Testing Spatial Error X

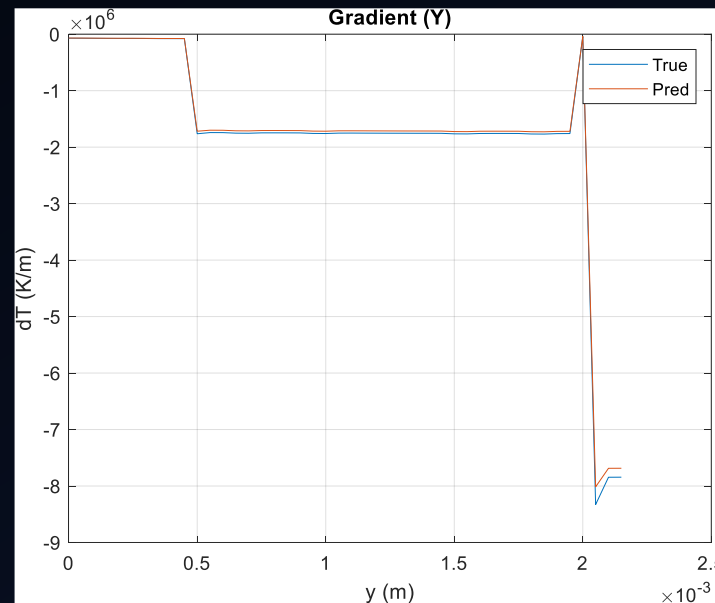
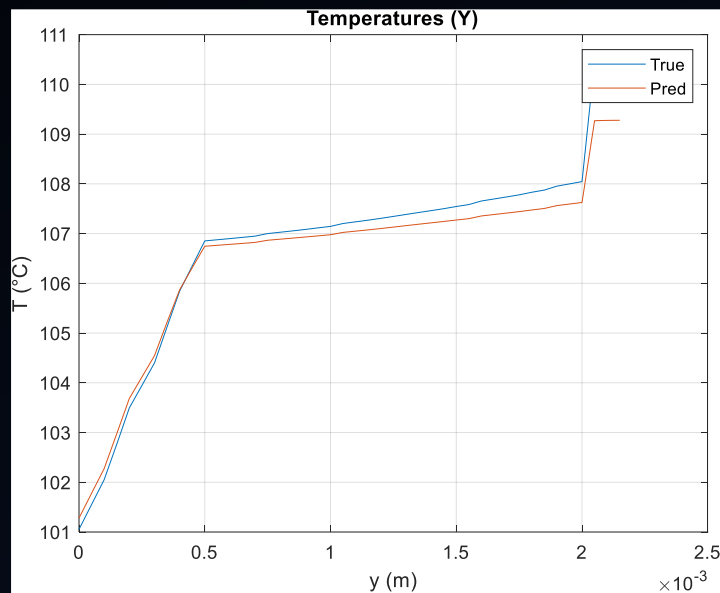
Literature:	-
Images:	-
Software:	TherMOS
Example:	tutorial_PO.m
Content:	Tutorial



- Both temperature and gradient are well captured
- Largest temperature error at boundary interface
- Largest gradient error at points of gradient changes

POD-Model: Testing Spatial Error Y

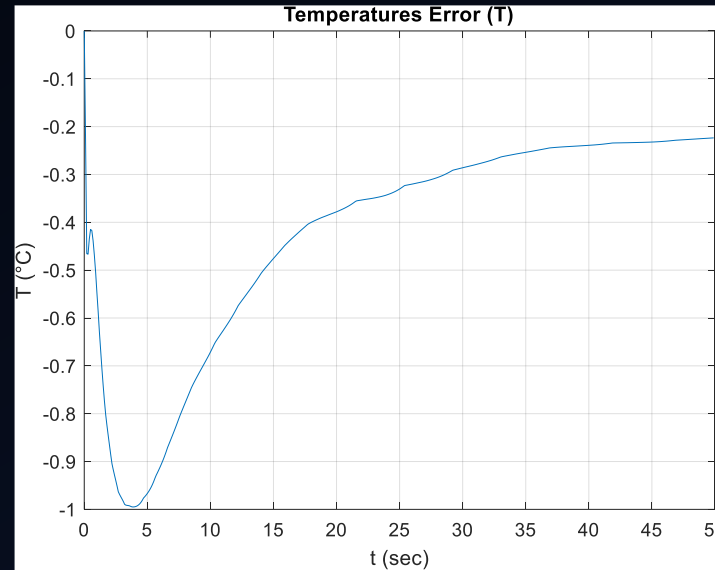
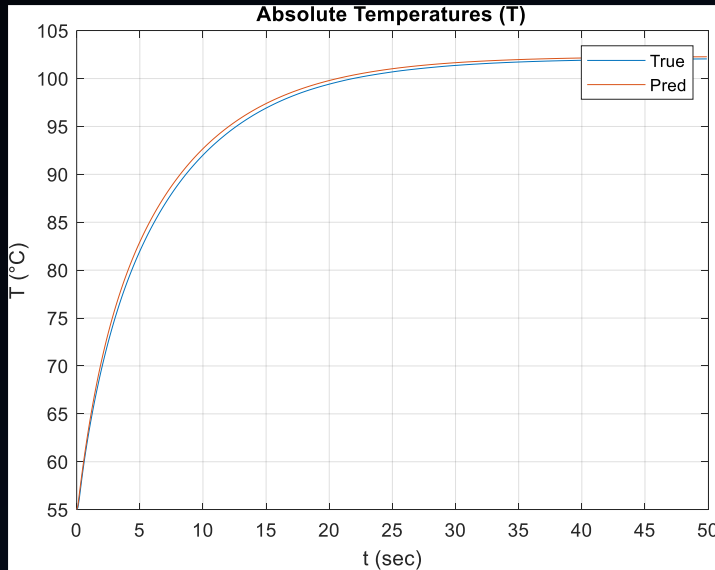
Literature:	-
Images:	-
Software:	TherMOS
Example:	tutorial_PO.m
Content:	Tutorial



- Gradient is directly proportional to the heat conductance of the PCB materials
- Temperature error increase in direction of the heat sources

POD-Model: Testing Temporal Error

Literature:	-
Images:	-
Software:	TherMOS
Example:	tutorial_PO.m
Content:	Tutorial



- Temporal behaviour is captured well
- Maximum error around 1,0 K
- Steady state error around 0,2 K



D3

DEEP LEARNING 1D MODEL ORDER REDUCTION



DL-CNN: Training and Testing Results

Literature: -
Images: -
Software: TherMOS
Example: tutorial_DL.m
Content: Tutorial

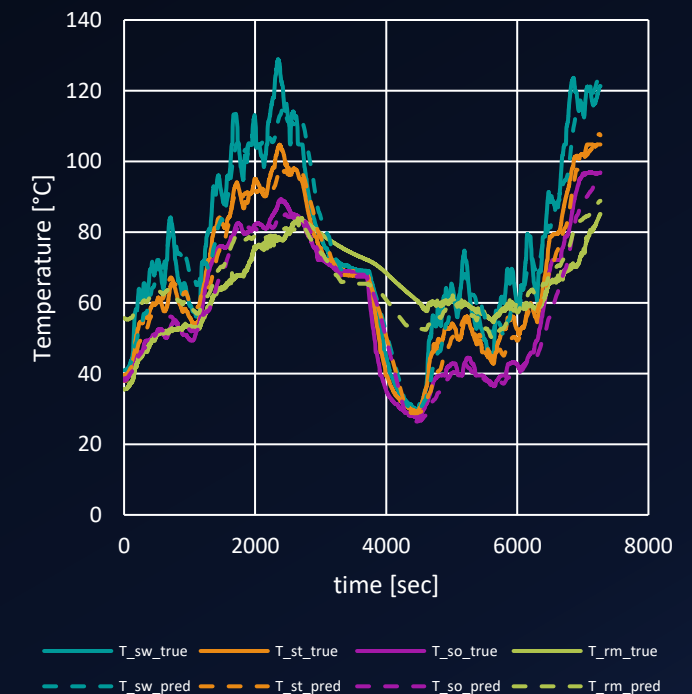
SETTINGS & PARAMETERS

- Normalisation X,y: off
- Learning rate: 0,001
- Batch: 1024
- Input window: 600
- Max Epochs: 100
- Layer Structure
 - 2x CNN ([5,64], [3,32])
 - 1x DNN (32 nodes)

STEADY & TRANS. ERROR METRIC

	MAE	MSE	MAX
T_sw	5,34	45,7	17,8
T_st	3,48	49,9	11,6
T_so	3,38	25,0	17,3
T_rm	6,12	53,0	20,2
Avg	4,58	35,9	20,2

TESTING DATA PLOT



DL-LSTM: Training and Testing Results

Literature: -
Images: -
Software: TherMOS
Example: tutorial_DL.m
Content: Tutorial

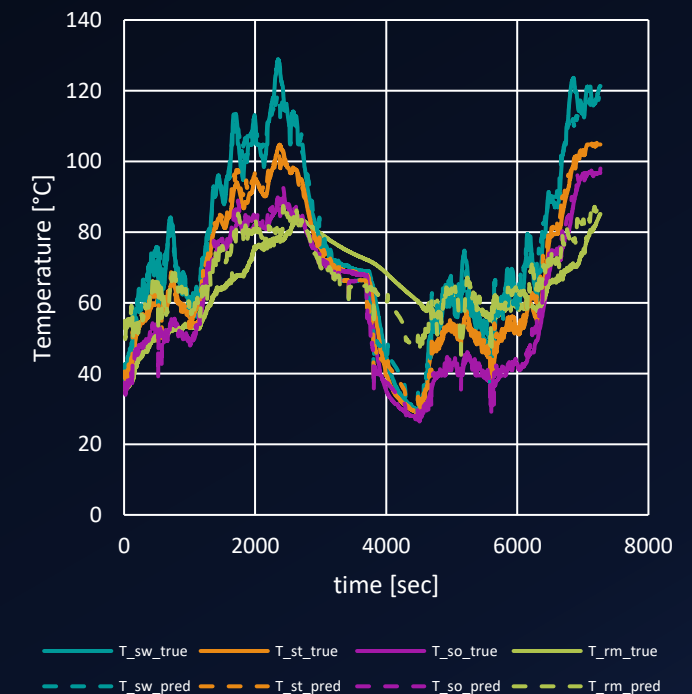
SETTINGS & PARAMETERS

- Normalisation X,y: off
- Learning rate: 0,001
- Batch: 1024
- Input window: 600
- Max Epochs: 100
- Layer Structure
 - 1x LSTM (64 nodes)
 - 1x DNN (64 nodes)

STEADY & TRANS. ERROR METRIC

	MAE	MSE	MAX
T_sw	3,29	18,4	26,9
T_st	2,23	9,37	22,4
T_so	1,92	6,47	17,0
T_rm	7,28	69,0	21,8
Avg	3,68	25,8	26,9

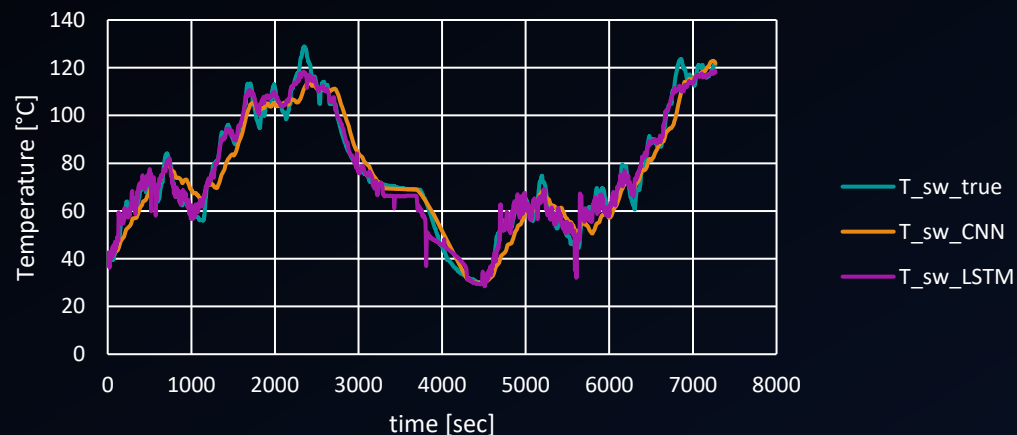
TESTING DATA PLOT



DL-Model Comparison

Literature:	-
Images:	-
Software:	TherMOS
Example:	tutorial_DL.m
Content:	Tutorial

ERROR COMPARISON STATOR WINDING (ID60)



	MAE	MSE	MAX
CNN	5,34	45,7	17,8
LSTM	3,29	18,4	26,9

DESCRIPTION & ANALYSIS

- LSTM outperforms CNN for MAE and MSE, but shows significantly higher MAX error
- CNN generate smoothing effect through convolution
- LSTM accurately models temporal behaviour



E

ADVANCED FUNCTIONALITIES





E1

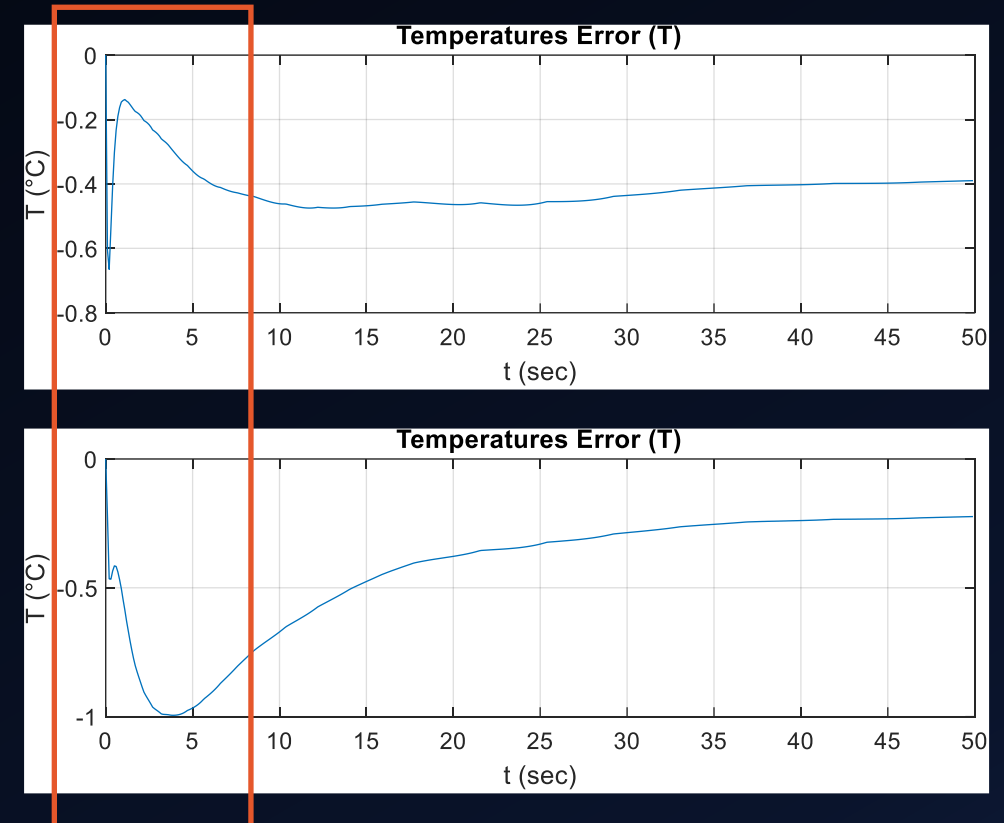
ADVANCED FUNCTIONALITIES: POD STATE SPACE MODEL



POD-SS Model

Literature:
Images:
Software:
Example:
Content:

- Instead of calculating C_{th} and G_{th} matrices, a state space system is used to model the temporal dependencies
- Improved transient modeling for low spatial and temporal resolution
- Max error reduces from -1,0 K to -0,65 K





E2

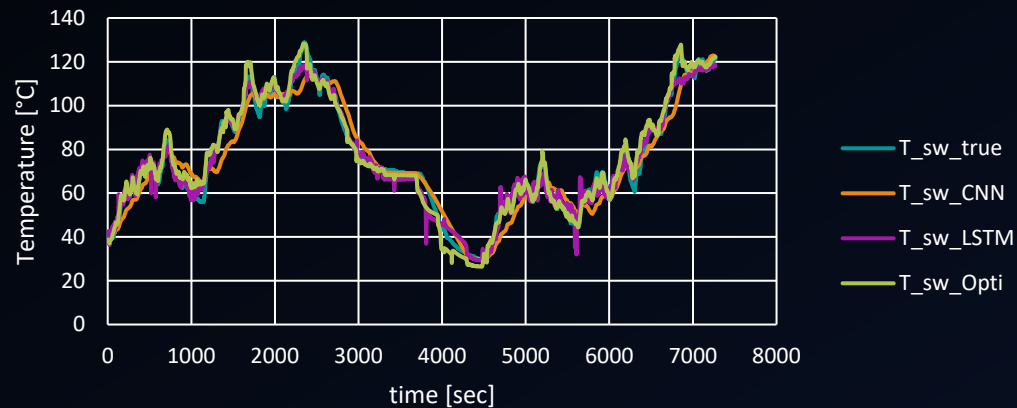
ADVANCED FUNCTIONALITIES: DEEP LEARNING



Optimised Neural Network Architecture

Literature:
Images:
Software:
Example:
Content:

ERROR COMPARISON STATOR WINDING (ID60)



	MAE	MSE	MAX
CNN	5,34	45,7	17,8
LSTM	3,29	18,4	26,9
Opti	2,80	13,4	13,5

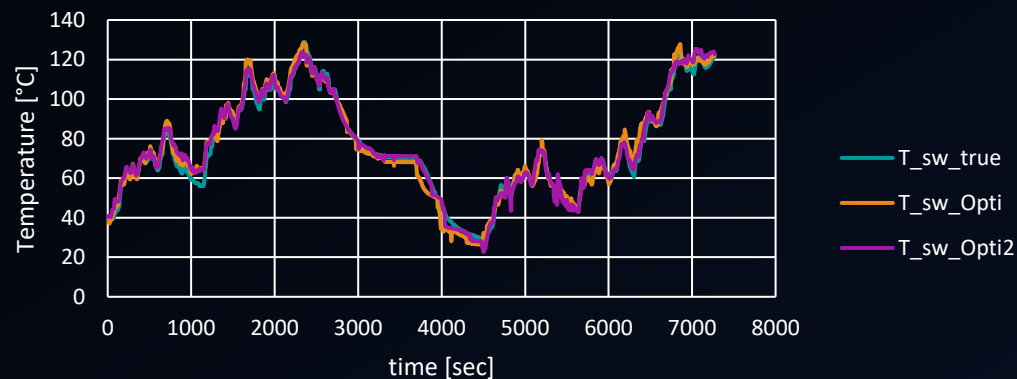
DESCRIPTION & ANALYSIS

- Combining a CNN and LSTM based architecture to utilise temporal and spatial convolution characteristics
- Significant further improvement for all error metric

Optimised Features and Neural Network Architecture

Literature:
Images:
Software:
Example:
Content:

ERROR COMPARISON STATOR WINDING (ID60)



	MAE	MSE	MAX
CNN	5,34	45,7	17,8
LSTM	3,29	18,4	26,9
Opti	2,80	13,4	13,5
Opti2	2,37	9,43	10,2

DESCRIPTION & ANALYSIS

- Adding additional input features, e.g. torque and rotational speed
- Further improvement in all error metric
- Lowest overall error



F

OUTLOOK AND CONCLUSION



Development and Future Goals

As failure and mistakes are inextricably linked to human nature, the toolkit is obviously not perfect, thus suggestions and constructive feedback are always welcome. If you want to contribute to the TherMOS toolkit or spotted any mistake, please contact me via: p.schirmer@herts.ac.uk. The following aspects will be improved in future version:

- 3D input files for the Proper Orthogonal Decomposition (POD)
- Physics Informed Neural Networks (PINNs) for 1D and 2D problems



G

REFERENCES



Bibliography

- [1] <https://www.ventec-group.com/products/tec-thermal-thermal-management-ims/vt-4b3/datasheet/>
- [2] W. Kirchgässner, O. Wallscheid and J. Böcker, "Estimating Electric Motor Temperatures With Deep Residual Machine Learning," in IEEE Transactions on Power Electronics, vol. 36, no. 7, pp. 7480-7488, July 2021, doi: 10.1109/TPEL.2020.3045596
- [3] <https://github.com/upb-lea/deep-pmsm>
- [4] <https://www.kaggle.com/datasets/wkirgsn/electric-motor-temperature>
- [5]